Kinora Knee

Clinical AI Orientation (Part 8)

Structured reference for RAG / AI-assisted physiotherapy consultation

Bones, Joints, Menisci, Ligaments, Muscles, Tendons, Bursae, Nerves, Vascular Supply, Biomechanics, Pathologies & Rehabilitation

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Disclaimer

This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established musculoskeletal concepts from standard anatomical texts (Gray's/Standring, Moore, Netter), kinesiology (Neumann), clinical assessment (Magee), sports medicine (Brukner & Khan), and specialty evidence themes (ACL RTS criteria, PFPS consensus, Cook & Purdam tendinopathy continuum, OARSI/NICE OA, LaPrade PLC). Approximate values vary by study. ***Red flags — knee dislocation (vascular emergency), septic arthritis, fracture, acute neurovascular deficit, or true locked knee — require urgent medical/orthopaedic referral.*** *Follow surgeon-specific protocols after operative care.*

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1. Knee Bones

RECORD: Femur (Distal)

ID: KNEE-BONE-01

Bone: Distal femur — medial and lateral condyles, intercondylar notch, trochlear groove

Region: Knee — distal thigh / tibiofemoral and patellofemoral articulations

Type: Long bone (distal epiphysis and metaphysis)

Landmarks: Medial/lateral femoral condyles; medial/lateral epicondyles; adductor tubercle; intercondylar notch (ACL/PCL); trochlear (patellar) groove; popliteal surface; sulcus terminalis

Articulations: Tibial plateaus via menisci (tibiofemoral); patella (patellofemoral)

Muscle Attachments: Gastrocnemius (posterior condyles); plantaris; popliteus (lateral condyle groove); articularis genu; adductor magnus to adductor tubercle; collateral ligaments near epicondyles

Ligament Attachments: ACL (posteromedial lateral femoral condyle / notch wall); PCL (anterolateral medial femoral condyle); MCL (medial epicondyle); LCL (lateral epicondyle); MPFL (near medial epicondyle / adductor tubercle complex)

Blood Supply: Genicular anastomosis (superior medial/lateral genicular, middle genicular into notch); descending genicular; popliteal branches

Venous Drainage: Genicular veins → popliteal vein

Innervation: Capsular/periosteal branches from femoral, tibial, common fibular, obturator (variable genicular nerves)

Biomechanical Role: Convex femoral condyles roll and glide on tibia; trochlea guides patella; notch houses cruciate ligaments

Load Transmission: Compressive load through condyles to tibia; medial compartment often higher load in neutral alignment; PF load through trochlea

Clinical Importance: OA preferential medial compartment; notch stenosis and ACL graft impingement; distal femur fractures; OCD of medial femoral condyle in youth

Common Fractures: Supracondylar; intercondylar (AO/OTA distal femur); osteochondral fractures with patellar dislocation

Healing Timeline: Metaphyseal fractures often 8–16 weeks depending on fixation/stability; cartilage injuries longer and variable

Imaging: AP/lateral/notch/sunrise radiographs; CT for fracture mapping; MRI for ligaments/cartilage/OCD

Rehabilitation Considerations: Protect fracture fixation WB orders; early ROM if stable; quad activation; alignment and gait retraining

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibia (Proximal)

ID: KNEE-BONE-02

Bone: Proximal tibia — medial/lateral plateaus, intercondylar eminence, tibial tuberosity

Region: Knee — proximal leg

Type: Long bone (proximal epiphysis/metaphysis)

Landmarks: Medial/lateral tibial plateaus; tibial spines (eminence); Gerdy's tubercle; tibial tuberosity; pes anserinus insertion area; posterior tibial slope

Articulations: Femoral condyles (via menisci); proximal fibula (proximal tibiofibular joint); patellar tendon at tuberosity

Muscle Attachments: Pes anserinus (sartorius, gracilis, semitendinosus); semimembranosus expansions; ITB/TFL to Gerdy's; tibialis anterior nearby; popliteus inserts posteromedial tibia

Ligament Attachments: ACL (anterior intercondylar area); PCL (posterior intercondylar area); meniscal roots/horns; MCL distal tibial attachment; coronary ligaments to plateau margins

Blood Supply: Inferior genicular arteries; anterior/posterior tibial recurrent; nutrient vessels

Venous Drainage: Genicular and tibial veins → popliteal

Innervation: Infrapatellar branch of saphenous; tibial and common fibular genicular branches

Biomechanical Role: Concave plateaus (lateral more convex) receive femoral load; slope influences ACL strain and anterior tibial translation

Load Transmission: Medial plateau larger/thicker cartilage typically; menisci expand contact area; tuberosity transmits extensor mechanism force

Clinical Importance: Tibial plateau fractures (Schatzker); ACL tibial footprint; Osgood-Schlatter at tuberosity; slope in ACL failure risk discussions

Common Fractures: Tibial plateau (lateral most common); tibial eminence avulsion (pediatric ACL equivalent); tuberosity avulsion

Healing Timeline: Plateau fractures often 12+ weeks; articular depression needs anatomic restoration for OA prevention

Imaging: AP/lateral/oblique; CT for plateau depression/split; MRI for soft tissue and occult fracture

Rehabilitation Considerations: WB per plateau depression and fixation; early ROM critical; progressive strength; monitor alignment

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Schatzker classification literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Fibula (Proximal)

ID: KNEE-BONE-03

Bone: Proximal fibula — fibular head and neck

Region: Lateral knee / proximal leg

Type: Long bone (proximal)

Landmarks: Fibular head; fibular neck (common fibular nerve winds around); styloid process (arcuate/PLC attachments)

Articulations: Proximal tibiofibular joint with lateral tibial condyle

Muscle Attachments: Biceps femoris to fibular head; soleus/peroneals more distal but relevant kinetic chain; fibularis longus origin nearby

Ligament Attachments: LCL to fibular head; popliteofibular ligament; arcuate complex / PLC structures; biceps femoris tendon continuity

Blood Supply: Inferior lateral genicular; anterior tibial recurrent; circumflex fibular branches

Venous Drainage: Accompanying veins to popliteal system

Innervation: Common fibular nerve closely related at neck — clinical landmark more than periosteal supply

Biomechanical Role: Lateral collateral and PLC anchor; proximal TFJ allows fibular rotation with ankle motion

Load Transmission: Minimal axial load vs tibia (~6–30% debated by study/condition); more ligamentous/lateral stabilizer role at knee

Clinical Importance: Common fibular nerve palsy at neck; Maisonneuve association; LCL/PLC injury with fibular head avulsion (arcuate sign)

Common Fractures: Fibular head/neck fracture; avulsion of styloid (arcuate sign on radiograph)

Healing Timeline: Typically 6–10 weeks if isolated; dictated by associated PLC/ankle injury

Imaging: AP/lateral knee including fibular head; MRI for PLC; assess nerve clinically

Rehabilitation Considerations: Screen foot drop; protect PLC repairs; progressive lateral stability training

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. PLC injury reviews (LaPrade).

RECORD: Patella

ID: KNEE-BONE-04

Bone: Patella — largest sesamoid bone

Region: Anterior knee within extensor mechanism

Type: Sesamoid bone

Landmarks: Base (superior); apex (inferior — patellar tendon); anterior subcutaneous surface; posterior medial/lateral facets; odd facet (far medial)

Articulations: Femoral trochlea (patellofemoral joint)

Muscle Attachments: Quadriceps tendon inserts on base/anterior; vastus medialis/lateralis expansions (retinacula); patellar tendon from apex

Ligament Attachments: MPFL to medial patella; LPFL/lateral retinaculum; medial/lateral patellotibial and patellomeniscal ligaments (variable)

Blood Supply: Genicular anastomosis (esp. inferior medial/lateral genicular; supreme/descending genicular) — bipartite and AVN considerations after surgery

Venous Drainage: Prepatellar venous plexus → genicular/popliteal veins

Innervation: Infrapatellar branch of saphenous; femoral cutaneous contributions — neuroma risk after incision

Biomechanical Role: Increases quadriceps moment arm; centralizes extensor force; thickest cartilage in body on posterior surface

Load Transmission: PF joint reaction force rises with flexion and quad demand — often several × BW in deep squat/stairs (study-dependent)

Clinical Importance: PFPS; patellar dislocation/instability; chondromalacia; bipartite patella; fracture; Sinding-Larsen-Johansson at apex in youth

Common Fractures: Transverse (most common); stellate; osteochondral with dislocation

Healing Timeline: ORIF/nonop 6–12 weeks typical bony union window; cartilage sequelae longer

Imaging: AP/lateral/sunrise (Merchant/Laurin); MRI for cartilage/MPFL; CT for fracture comminution

Rehabilitation Considerations: Protect extensor mechanism repairs; progressive ROM; VMO/quad control without early aggressive deep loaded flexion if chondral injury

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. PF biomechanics reviews.

2. Joints

RECORD: Tibiofemoral Joint

ID: KNEE-JNT-01

Joint: Tibiofemoral joint

Type: Modified hinge (bicondylar synovial) — flexion/extension with coupled rotation

Capsule: Fibrous capsule reinforced by collaterals, retinacula, oblique popliteal and arcuate ligaments; synovial lining with multiple recesses

Articular Cartilage: Hyaline cartilage on femoral condyles and tibial plateaus; thickness varies; menisci interposed

Ligaments: ACL, PCL, MCL, LCL; posteromedial and posterolateral corner complexes; coronary ligaments to menisci

Menisci: Medial and lateral menisci deepen tibial surface and distribute load

Blood Supply: Genicular anastomosis; middle genicular to cruciates

Innervation: Posterior articular nerve (tibial); genicular branches of femoral, common fibular, obturator — rich nociception/proprioception

Arthrokinematics: Flexion: femoral condyles roll posterior and glide anterior on tibia (open chain opposite described for tibia-on-femur). Screw-home: tibia ER on femur into terminal extension (open chain)

Osteokinematics: Flexion/extension primary; IR/ER increases with flexion; abduction/adduction minimal (collateral constrained)

Normal ROM: Extension 0 to slight hyperextension (~5 deg); flexion ~135–150 deg (soft tissue dependent); rotation ~20–30 deg at 90 deg flexion (approx.)

Joint Reaction Forces: Compressive forces often 2–4× BW walking and higher in running/squatting (instrumented and model estimates vary)

Clinical Assessment: Effusion (stroke/ballottement); alignment (Q-angle, varus/valgus); ROM; strength; ligament laxity; meniscal signs; gait

Special Tests: Lachman, anterior/posterior drawer, pivot shift, valgus/varus stress, McMurray, Thessaly, dial test — interpret in cluster

Common Pathologies: Ligament tears, meniscal injury, OA, OCD, loose bodies, arthrofibrosis

Imaging: Radiographs first for fracture/OA; MRI for soft tissue; CT for complex fracture

Manual Therapy: Tibiofemoral AP/PA glides, distraction, soft-tissue — as adjunct within pain-free ranges; avoid aggressive mobility on acute ligament tears

Rehabilitation: Quad activation, ROM restoration, progressive strengthening, neuromuscular control, criteria-based RTS

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Patellofemoral Joint

ID: KNEE-JNT-02

Joint: Patellofemoral joint

Type: Synovial plane/saddle-like articulation between patella and trochlea

Capsule: Continuous with knee capsule; medial/lateral retinacula; suprapatellar pouch communication

Articular Cartilage: Thick patellar cartilage; trochlear cartilage — early PF OA common

Ligaments: MPFL primary medial stabilizer 0–30 deg flexion; medial/lateral retinacula; LPFL less discrete

Menisci: Not primary PF structures; patellomeniscal ligaments variable

Blood Supply: Genicular anastomosis around patella

Innervation: Infrapatellar saphenous branch; retinacular nerves — anterior knee pain substrate

Arthrokinematics: Patella glides inferior with flexion and superior with extension; medial-lateral tilt/rotation accompany tracking

Osteokinematics: Patellar motion coupled to tibiofemoral flexion angle; contact area migrates proximal on patella with deeper flexion

Normal ROM: Patellar mobility assessed mediolateral glide (~1–2 quadrants typical teaching — individualize); TF flexion ROM governs PF contact

Joint Reaction Forces: Increase with knee flexion angle and quad force; stairs and deep squat high PF load

Clinical Assessment: Tracking, apprehension, grind/clarke (interpret cautiously), step-down quality, hip/foot contributors

Special Tests: Patellar apprehension; moving apprehension; J-sign observation; compression tests limited specificity

Common Pathologies: PFPS, instability/dislocation, chondral lesions, OA, plica irritation

Imaging: Sunrise/Merchant views; MRI for MPFL/cartilage; TT-TG distance in instability workups

Manual Therapy: Medial patellar glides if lateral tightness; soft-tissue ITB/lateral retinaculum — evidence mixed; combine with exercise

Rehabilitation: Quad strengthening (including VMO emphasis as part of quad, not isolated myth); hip abductor/ER; load management; taping/orthoses adjunct

References: Neumann DA. Kinesiology of the Musculoskeletal System. PFPS consensus statements. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Proximal Tibiofibular Joint

ID: KNEE-JNT-03

Joint: Proximal tibiofibular joint (PTFJ)

Type: Plane synovial joint

Capsule: Fibrous capsule with anterior and posterior tibiofibular ligaments

Articular Cartilage: Hyaline on tibial facet and fibular head

Ligaments: Anterior and posterior proximal tibiofibular ligaments; association with LCL/biceps complex

Menisci: None

Blood Supply: Inferior lateral genicular; anterior tibial recurrent

Innervation: Common fibular nerve branches; recurrent genicular

Arthrokinematics: Small superior/inferior and rotational glides with ankle dorsi/plantarflexion

Osteokinematics: Accessory motion supporting ankle; not a primary knee mover

Normal ROM: Minimal — assessed by fibular head AP glide

Joint Reaction Forces: Low compared with tibiofemoral; higher with certain ankle-loaded sports

Clinical Assessment: Fibular head tenderness/mobility; common fibular nerve screen; differentiate LCL/PLC pain

Special Tests: Fibular head translation; ankle dorsiflexion with PTFJ symptom change

Common Pathologies: Instability (anterolateral dislocation rare); OA; referred pain; injury with PLC

Imaging: Radiographs comparing sides; MRI if PLC/nerve concern

Manual Therapy: Fibular head AP/PA mobilizations when hypomobile and indicated

Rehabilitation: Progressive loading; address ankle DF; lateral knee stability if PLC involved

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

3. Menisci

RECORD: Medial Meniscus

ID: KNEE-MEN-01

Structure: C-shaped fibrocartilage disc; broader posterior horn; firmly attached to MCL deep layer / capsule

Region: Medial tibiofemoral compartment

Attachments: Anterior and posterior roots to tibial intercondylar region; coronary (meniscotibial) ligaments; deep MCL; semimembranosus expansions posteriorly

Blood Supply: Peripheral capillary plexus from genicular/middle genicular — red-red outer third; inner two-thirds relatively avascular

Innervation: Peripheral horns richly innervated (mechanoreceptors/nociceptors); free edge less

Zones: Red-red (peripheral), red-white (middle), white-white (central) — healing potential follows vascularity

Function: Load distribution, shock absorption, joint stability (esp. anterior translation secondary), lubrication/nutrition

Shock Absorption: Converts compressive loads to hoop stresses via circumferential collagen

Load Distribution: Increases contact area; medial meniscectomy substantially elevates peak contact stress

Biomechanics: Less mobile than lateral (~2–5 mm translation); torn more often partly due to MCL fixation

Clinical Tests: McMurray (medial), Thessaly, Apley compression, joint-line tenderness — clusters better than single tests

MRI Findings: Intrameniscal signal to articular surface (Grade 3 tear); root tear with extrusion; ramp lesions with ACL

Healing Potential: Peripheral longitudinal tears best; white-white poor; roots critical if repaired early

Common Injuries: Posterior horn tears; bucket-handle; root tears; degenerative horizontal cleavage in middle age

Mechanism: Twisting on flexed planted foot; degenerative attrition; ACL injury association (medial ramp)

Conservative Treatment: Activity modification, effusion control, quad strength, progressive loading for stable degenerative tears per evidence

Surgical Treatment: Repair preferred when feasible (esp. young/peripheral/root); partial meniscectomy if irreparable; meniscal transplant selected cases

Rehabilitation: Repair: protected WB/ROM per protocol often 4–6+ weeks. Meniscectomy: faster ROM/WB progression

Return to Sport: Repair often 4–6 months criteria-based; meniscectomy earlier if strength/effusion allow — individualize

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Meniscal repair healing literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Beaufils/Pujol meniscus guidelines themes.

RECORD: Lateral Meniscus

ID: KNEE-MEN-02

Structure: More circular O-shaped fibrocartilage; covers greater % of lateral plateau; more mobile

Region: Lateral tibiofemoral compartment

Attachments: Anterior/posterior roots; loose coronary ligaments; no firm LCL attachment; meniscofemoral ligaments (Humphrey/Wrisberg) variable to PCL

Blood Supply: Similar peripheral-to-central gradient from genicular plexus

Innervation: Peripheral mechanoreceptors/nociceptors

Zones: Red-red / red-white / white-white as medial

Function: Lateral load distribution; rotational congruence; secondary stability

Shock Absorption: Hoop stress mechanism; discoid variants alter mechanics

Load Distribution: Lateral meniscectomy particularly detrimental to contact mechanics

Biomechanics: Translation ~9–11 mm possible — more mobile, slightly lower tear rate than medial overall but common with ACL

Clinical Tests: McMurray lateral; Thessaly; joint-line tenderness; bounce home

MRI Findings: Tear patterns; discoid meniscus; meniscofemoral ligament mimics; root/extrusion

Healing Potential: Vascular zone dependent; discoid saucerization considerations in youth

Common Injuries: ACL-associated lateral tears; discoid tear in children; popliteomeniscal fascicle injuries

Mechanism: Pivot-shift ACL mechanism; squat/twist; discoid abnormal morphology

Conservative Treatment: Selected stable tears; activity and strength programs

Surgical Treatment: Repair when possible; saucerization of symptomatic discoid; preserve tissue

Rehabilitation: Similar repair protections; emphasize quad and neuromuscular control

Return to Sport: Criteria-based; concurrent ACL dictates timeline often

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Discoid meniscus reviews.

4. Ligaments

RECORD: ACL

ID: KNEE-LIG-01

Ligament: Anterior cruciate ligament

Origin: Posteromedial aspect of lateral femoral condyle (intercondylar notch wall)

Insertion: Anterior intercondylar area of tibia, just anterior/between tibial spines

Primary Function: Resists anterior tibial translation; resists tibial IR; key rotational stabilizer

Secondary Function: Secondary restraint to varus/valgus at extension; proprioception

Biomechanics: Anteromedial and posterolateral bundles; AM tighter in flexion, PL in extension (simplified teaching model)

Load Capacity: Ultimate load classically about 1700-2200 N in young specimens (study-dependent); grafts remodel over months

Blood Supply: Middle genicular artery primary; fat pad contributions

Healing: Intra-articular — poor spontaneous healing of complete tears; reconstruction common in cutting athletes

Clinical Tests: Lachman (most sensitive), anterior drawer, pivot shift (specific for rotational laxity)

Common Injuries: Noncontact pivot (cut/land); contact valgus; often with meniscus/MCL (unhappy triad)

MRI Findings: Fiber discontinuity, abnormal signal, bone bruises (lateral femoral condyle + posterior lateral tibia classic)

Ultrasound: Limited for ACL proper; not first-line

Treatment: Coper vs reconstruction shared decision; bracing/rehab for selected; ACLR for instability/sport demand

Rehabilitation: Quad activation, extension early, progressive strength/NMC, criteria-based RTS often 9-12+ months

Return to Sport: Time + criteria (strength symmetry, hop tests, psych readiness); reinjury risk if premature

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. AAOS/AOSSM ACL themes. Musahl/Karlsson consensus literature.

RECORD: PCL

ID: KNEE-LIG-02

Ligament: Posterior cruciate ligament

Origin: Anterolateral medial femoral condyle (notch)

Insertion: Posterior tibial sulcus / posterior intercondylar area

Primary Function: Resists posterior tibial translation

Secondary Function: Secondary rotational and varus/valgus restraint

Biomechanics: AL and PM bundles; strongest knee ligament by cross-section

Load Capacity: Higher ultimate load than ACL in many cadaver series (approx 2500-4000 N — study-dependent)

Blood Supply: Middle genicular artery

Healing: Better healing potential than ACL; many isolated PCL injuries treated nonoperatively

Clinical Tests: Posterior drawer, posterior sag (Godfrey), quadriceps active test; dial if PLC combined

Common Injuries: Dashboard injury; fall on flexed knee with plantarflexed foot; combined PLC

MRI Findings: Fiber disruption/thickening; posterior tibial translation signs

Ultrasound: Not primary

Treatment: Grade I-II often conservative; grade III / combined / bony avulsion may need surgery

Rehabilitation: Avoid early isolated hamstring loading that posteriorly shears; progressive quad; protect against posterior sag

Return to Sport: Often 3-9 months depending on grade/surgery; criteria-based

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. PCL management reviews.

RECORD: MCL

ID: KNEE-LIG-03

Ligament: Medial collateral ligament (tibial collateral) — superficial and deep layers

Origin: Medial femoral epicondyle

Insertion: Superficial: proximal medial tibia about 5-7 cm below joint. Deep: meniscocapsular attachments

Primary Function: Primary restraint to valgus; resists tibial ER

Secondary Function: Secondary anterior translation restraint (especially if ACL deficient)

Biomechanics: Superficial MCL primary valgus stabilizer across flexion; deep MCL linked to medial meniscus

Load Capacity: Robust healing potential compared with ACL; grades I-II usually heal

Blood Supply: Inferior medial genicular and soft-tissue plexus — relatively good

Healing: Extra-articular — excellent healing with protected motion for most grade I-II

Clinical Tests: Valgus stress at 0 and 30 degrees; 30 degrees isolates MCL more

Common Injuries: Valgus blow; cutting; unhappy triad with ACL/medial meniscus

MRI Findings: Fiber edema/discontinuity; femoral insertion common; Pellegrini-Stieda chronic calcification

Ultrasound: Useful for superficial MCL fiber disruption and dynamic gapping

Treatment: Bracing and early ROM for most; surgery rare (distal Stener-like / multi-ligament)

Rehabilitation: Early motion in brace; progressive adductor/quad; avoid early aggressive valgus stress

Return to Sport: Grade I about 1-3 wks; II about 3-6 wks; III often 6-12+ wks — criteria-based

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. MCL healing literature.

RECORD: LCL

ID: KNEE-LIG-04

Ligament: Lateral collateral ligament (fibular collateral)

Origin: Lateral femoral epicondyle

Insertion: Fibular head

Primary Function: Primary restraint to varus at 30 degrees flexion

Secondary Function: Secondary ER/rotation restraint with PLC

Biomechanics: Cord-like; extracapsular; distinct from popliteus tendon

Load Capacity: Smaller than MCL; often injured with PLC not in isolation

Blood Supply: Inferior lateral genicular / soft tissue

Healing: Variable; grade III / PLC often need surgical consideration

Clinical Tests: Varus stress 0 and 30 degrees; dial test; posterolateral drawer

Common Injuries: Varus trauma; combined PLC; fibular head avulsion

MRI Findings: LCL discontinuity; associated PLC/biceps; arcuate sign on X-ray if styloid avulsion

Ultrasound: Can visualize LCL dynamically

Treatment: Isolated low-grade conservative; unstable PLC/LCL often reconstruct

Rehabilitation: Protect varus; progressive lateral strength; gait with unloader strategies if needed

Return to Sport: Depends on PLC involvement — often months after reconstruction

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. LaPrade PLC/LCL literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: MPFL

ID: KNEE-LIG-05

Ligament: Medial patellofemoral ligament

Origin: Between adductor tubercle and medial epicondyle (Schottle point surgical landmark)

Insertion: Upper/medial patella (proximal half typically)

Primary Function: Primary soft-tissue restraint to lateral patellar dislocation in early flexion (0-30 degrees)

Secondary Function: Guides patellar engagement into trochlea

Biomechanics: Highest contribution to medial restraint near extension; trochlea bony constraint dominates deeper flexion

Load Capacity: Fails commonly in first-time lateral dislocation

Blood Supply: Medial genicular / soft-tissue network

Healing: May heal elongated; recurrent instability if anatomic risk factors (trochlear dysplasia, TT-TG, patella alta)

Clinical Tests: Patellar apprehension; moving apprehension; lateral glide excess

Common Injuries: Lateral patellar dislocation — MPFL tear often at femoral or patellar attachment

MRI Findings: MPFL disruption; medial patellar / lateral femoral osteochondral injury; bone bruise pattern

Ultrasound: Dynamic MPFL/medial restraint assessment adjunct

Treatment: First dislocation often conservative if reduced and no unstable osteochondral fragments; reconstruction for recurrent

Rehabilitation: Brace/ROM protocols; VMO/quad and hip strength; avoid early provocative lateral stress

Return to Sport: Often 3-6+ months criteria-based after reconstruction

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Patellar instability consensus literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Posterolateral Corner

ID: KNEE-LIG-06

Ligament: Posterolateral corner complex (LCL, popliteus tendon, popliteofibular ligament, capsule/arcuate)

Origin: Lateral femoral condyle / fibular head / posterolateral tibia (structure-specific)

Insertion: Structure-specific — see LCL, popliteus, PFL

Primary Function: Resists varus, tibial ER, and posterior translation near extension

Secondary Function: Protects ACL/PCL from abnormal coupled motions

Biomechanics: Critical in combined ligament injury; missed PLC raises ACL/PCL graft failure risk

Load Capacity: Complex failure under high-energy trauma

Blood Supply: Inferior lateral genicular and popliteal branches

Healing: Grade III PLC often poor with nonoperative care alone — early diagnosis important

Clinical Tests: Dial at 30/90, posterolateral drawer, reverse pivot, varus stress, external rotation recurvatum

Common Injuries: High-energy varus/hyperextension/twist; combined ACL/PCL

MRI Findings: LCL/popliteus/PFL injury; bone bruises; arcuate sign

Ultrasound: Limited completeness vs MRI

Treatment: Acute repair/reconstruct per injury pattern; chronic reconstruction

Rehabilitation: Protected ROM/WB per surgery; progressive rotational control; delayed aggressive pivoting

Return to Sport: Often 9-12+ months in multi-ligament knees

References: LaPrade RF PLC anatomy/reconstruction. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Posteromedial Corner

ID: KNEE-LIG-07

Ligament: Posteromedial corner (posterior oblique ligament, semimembranosus expansions, posteromedial capsule, deep MCL)

Origin: Medial femoral epicondyle / adductor tubercle region and posteromedial tibia

Insertion: Posteromedial tibia / meniscocapsular complex

Primary Function: Resists valgus and tibial IR/anteromedial rotational instability near extension

Secondary Function: Supports medial meniscus; complements superficial MCL

Biomechanics: POL tight in extension; important in AMRI with ACL injury

Load Capacity: Often injured with MCL/ACL

Blood Supply: Medial genicular network

Healing: Capsular structures may heal with protected motion; residual AMRI may need addressing

Clinical Tests: Valgus at 0 degrees (capsule/PMC); anteromedial drawer; Slocum test variants

Common Injuries: Valgus-rotation; ACL + medial sided injury

MRI Findings: POL/capsular edema; meniscocapsular separation (ramp)

Ultrasound: Adjunct for MCL; PMC deeper harder

Treatment: Often with MCL bracing; surgery in multi-ligament / persistent AMRI

Rehabilitation: Early protected ROM; quad; progressive rotational control

Return to Sport: Tied to MCL/ACL timelines

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. PMC/AMRI literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Coronary Ligaments

ID: KNEE-LIG-08

Ligament: Coronary (meniscotibial) ligaments

Origin: Peripheral meniscal margins

Insertion: Tibial plateau margins

Primary Function: Anchor menisci to tibia; allow controlled meniscal motion

Secondary Function: Contribute to meniscal stability and load sharing

Biomechanics: Disruption permits meniscal extrusion and altered hoop stress

Load Capacity: Can tear with meniscal root/peripheral injuries

Blood Supply: Peripheral meniscal capillary plexus

Healing: Peripheral — potential if vascular zone involved and stabilized

Clinical Tests: No isolated bedside test; inferred with meniscal exam and MRI

Common Injuries: With meniscal tears, root injuries, trauma

MRI Findings: Meniscal extrusion; meniscotibial ligament discontinuity

Ultrasound: Limited

Treatment: Address with meniscal repair when indicated

Rehabilitation: Per meniscal repair protocol

Return to Sport: Per associated meniscal surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Meniscotibial ligament / extrusion literature.

5. Muscles

RECORD: Rectus Femoris

ID: KNEE-MUS-01

Muscle: Rectus femoris

Origin: AIIS (straight head) and superior acetabular rim (reflected head)

Insertion: Quadriceps tendon to patella to tibial tuberosity via patellar tendon

Innervation: Femoral nerve (L2-L4)

Blood Supply: LCFA and femoral branches

Fiber Direction: Vertical biarticular anterior thigh

Primary Action: Knee extension and hip flexion

Secondary Action: Anterior pelvic tilt influence

Stabilizing Function: Extensor mechanism; decelerates knee flexion

Synergists: Vastii

Antagonists: Hamstrings

Length-Tension Relationship: Active insufficiency with combined hip flexion + knee extension; passive insufficiency opposite

EMG Activation: High in kicking, sprinting, jumping

Trigger Points: Anterior thigh

Pain Referral: Anterior thigh to knee

Stretch: Prone quad stretch with hip extension

Strengthening: Short-arc quads, Spanish squat, graded kicking

Neuromuscular Control: Coordinate hip-knee coupling in landing

Clinical Importance: AIIS apophysitis; RF strain in kickers

Common Injuries: Myotendinous strain; AIIS avulsion (youth)

Rehabilitation: Progressive loading; delayed max kicking

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Vastus Medialis

ID: KNEE-MUS-02

Muscle: Vastus medialis (including VMO oblique fibers)

Origin: Medial linea aspera, intertrochanteric line, medial supracondylar line, tendons of adductors

Insertion: Medial quad tendon / medial patellar retinaculum / patella

Innervation: Femoral nerve (L2-L4)

Blood Supply: Femoral / descending genicular

Fiber Direction: Oblique distal fibers (VMO) medially to patella

Primary Action: Knee extension; medial patellar stabilization (VMO contribution)

Secondary Action: Last-degree extension contribution debated

Stabilizing Function: Medial PF tracking restraint with MPFL

Synergists: Other vastii, RF

Antagonists: Hamstrings

Length-Tension Relationship: Best force mid-range extension; atrophy common after effusion/ACL

EMG Activation: Active in terminal extension and SLR; open vs closed chain both useful

Trigger Points: Medial thigh to medial knee

Pain Referral: Medial knee

Stretch: Not typically stretched in isolation

Strengthening: Quad sets, TKE, step-downs, closed-chain squats

Neuromuscular Control: Timing with VL for tracking; address effusion first

Clinical Importance: PFPS; post-op inhibition (arthrogenic muscle inhibition)

Common Injuries: AMI after injury/surgery

Rehabilitation: NMES adjunct; effusion control; progressive closed chain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Vastus Lateralis

ID: KNEE-MUS-03

Muscle: Vastus lateralis

Origin: Greater trochanter, linea aspera, intertrochanteric line, lateral intermuscular septum

Insertion: Lateral quad tendon / lateral retinaculum / patella

Innervation: Femoral nerve (L2-L4)

Blood Supply: LCFA descending branch

Fiber Direction: Large lateral bulk — lateral pull on patella

Primary Action: Knee extension

Secondary Action: Lateral patellar force vector

Stabilizing Function: Extensor torque; can dominate if medial structures weak/tight lateral tissues

Synergists: Other vastii, RF

Antagonists: Hamstrings

Length-Tension Relationship: Large CSA — major extensor torque contributor

EMG Activation: High in squats, lunges, jumping

Trigger Points: Lateral thigh

Pain Referral: Lateral thigh to lateral knee

Stretch: ITB/lateral retinacular soft-tissue techniques as adjunct

Strengthening: Squats, lunges, step-ups

Neuromuscular Control: Balance with medial quad and hip control for PF tracking

Clinical Importance: PFPS lateral overload patterns; ITB interplay

Common Injuries: Contusion; strain less than RF

Rehabilitation: Load management; hip abductor strength; avoid chronic lateral tightness assumptions alone

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Vastus Intermedius

ID: KNEE-MUS-04

Muscle: Vastus intermedius

Origin: Anterior and lateral femoral shaft

Insertion: Deep quad tendon / patella

Innervation: Femoral nerve (L2-L4)

Blood Supply: LCFA / femoral perforators

Fiber Direction: Deep vertical under RF

Primary Action: Knee extension

Secondary Action: Articularis genu related to synovial retraction

Stabilizing Function: Deep extensor force

Synergists: Other vastii

Antagonists: Hamstrings

Length-Tension Relationship: Works across knee ROM with vastii group

EMG Activation: Co-activates with quad set/squat

Trigger Points: Deep anterior thigh

Pain Referral: Anterior thigh

Stretch: With RF stretch

Strengthening: Same quad progressions

Neuromuscular Control: Part of global quad reactivation after AMI

Clinical Importance: Often overlooked contributor to extension lag

Common Injuries: Contusion deep thigh

Rehabilitation: Progressive quad loading post-injury

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Biceps Femoris

ID: KNEE-MUS-05

Muscle: Biceps femoris (long and short heads)

Origin: LH: ischial tuberosity. SH: linea aspera / lateral supracondylar line

Insertion: Fibular head (and lateral tibial contributions)

Innervation: LH tibial nerve; SH common fibular nerve (L5-S2)

Blood Supply: Perforators / inferior gluteal

Fiber Direction: Lateral posterior thigh — biarticular LH

Primary Action: Knee flexion and tibial ER; LH hip extension

Secondary Action: Hip extension (long head); assists PLC dynamically

Stabilizing Function: PLC dynamic support with LCL/biceps complex

Synergists: Other hamstrings

Antagonists: Quadriceps

Length-Tension Relationship: High late-swing sprint strain risk when lengthening

EMG Activation: Very high late-swing sprint EMG (LH)

Trigger Points: Lateral posterior thigh

Pain Referral: To fibular head / lateral knee

Stretch: Supine SLR-based stretch with neural differentiation

Strengthening: Nordics, RDLs, leg curls, sprint drills

Neuromuscular Control: Deceleration control; fibular head stability with LCL

Clinical Importance: Most common sprint hamstring strain (LH); PLC association at insertion

Common Injuries: Myofascial strain; distal insertion injury

Rehabilitation: BAMIC-informed progressive lengthening rehab

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Semitendinosus

ID: KNEE-MUS-06

Muscle: Semitendinosus

Origin: Ischial tuberosity (conjoint with BF LH)

Insertion: Pes anserinus (proximal medial tibia)

Innervation: Tibial nerve (L5-S2)

Blood Supply: Perforators

Fiber Direction: Medial posterior — long tendon

Primary Action: Knee flexion and tibial IR; hip extension

Secondary Action: Pes anserinus medial stability

Stabilizing Function: Dynamic MCL assistant

Synergists: SM, BF

Antagonists: Quadriceps

Length-Tension Relationship: Common ACL autograft harvest — monitor donor morbidity

EMG Activation: High in Nordic and RDL

Trigger Points: Medial posterior thigh

Pain Referral: Medial knee / pes

Stretch: Hamstring stretch

Strengthening: Nordics, curls, RDL

Neuromuscular Control: Medial chain control in cutting

Clinical Importance: Graft harvest; proximal tendinopathy; pes bursitis differential

Common Injuries: Strain; harvest-related weakness

Rehabilitation: Progressive eccentric; graft-site rehab after ACLR

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Semimembranosus

ID: KNEE-MUS-07

Muscle: Semimembranosus

Origin: Ischial tuberosity (separate footprint)

Insertion: Posteromedial tibial condyle with expansions to PMC/meniscus

Innervation: Tibial nerve (L5-S2)

Blood Supply: Perforators

Fiber Direction: Deep medial hamstring

Primary Action: Knee flexion, tibial IR, hip extension

Secondary Action: Retracts medial meniscus posteriorly; PMC tension

Stabilizing Function: Posteromedial stability

Synergists: ST, BF

Antagonists: Quadriceps

Length-Tension Relationship: Expansions clinically important in PMC and Baker cyst region

EMG Activation: High in hip hinge and curl

Trigger Points: Posteromedial knee

Pain Referral: Posteromedial knee

Stretch: Hamstring stretch

Strengthening: RDL, curls, bridge progressions

Neuromuscular Control: PMC control with ACL rehab

Clinical Importance: PMC injury; distal tendinopathy; Baker cyst association

Common Injuries: Distal insertional pain; strain

Rehabilitation: Load progressive; differentiate meniscal pain

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Popliteus

ID: KNEE-MUS-08

Muscle: Popliteus

Origin: Lateral femoral condyle (intra-articular origin) and fibular head via popliteofibular ligament complex

Insertion: Posterior proximal tibia above soleal line

Innervation: Tibial nerve (L4-S1)

Blood Supply: Inferior medial genicular / popliteal branches

Fiber Direction: Oblique across posterior knee

Primary Action: Unlocks knee by tibial IR (open chain) / femoral ER (closed chain) initiating flexion

Secondary Action: Assists knee flexion; resists tibial ER

Stabilizing Function: Dynamic PLC stabilizer

Synergists: LCL / popliteofibular ligament complex

Antagonists: Knee extensors; tibial external rotators

Length-Tension Relationship: Critical in screw-home reversal

EMG Activation: Active at flexion initiation and rotational control

Trigger Points: Posterior knee

Pain Referral: Posterolateral knee

Stretch: Gentle tibial ER stretch variants

Strengthening: Tibial IR against resistance; terminal stance control drills

Neuromuscular Control: PLC rehab rotational control

Clinical Importance: PLC injury; difficult isolated strain diagnosis

Common Injuries: With PLC trauma

Rehabilitation: Protect varus/ER early after PLC surgery

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Gastrocnemius

ID: KNEE-MUS-09

Muscle: Gastrocnemius (medial and lateral heads)

Origin: Posterior femoral condyles

Insertion: Achilles tendon to calcaneus

Innervation: Tibial nerve (S1-S2)

Blood Supply: Sural branches of popliteal

Fiber Direction: Biarticular posterior — crosses knee and ankle

Primary Action: Ankle plantarflexion; knee flexion assist

Secondary Action: Posterior capsular dynamic support

Stabilizing Function: Resists knee hyperextension dynamically

Synergists: Soleus (ankle), hamstrings (knee)

Antagonists: Tibialis anterior; quadriceps

Length-Tension Relationship: Active insufficiency with knee flexion + PF

EMG Activation: High in sprint push-off and jump

Trigger Points: Calf to posterior knee

Pain Referral: Posterior knee / calf — DVT differential important

Stretch: Wall gastroc stretch knee extended

Strengthening: Heel raises (knee extended biases gastroc)

Neuromuscular Control: Deceleration and landing stiffness control

Clinical Importance: Strain (tennis leg medial head); Baker cyst differential

Common Injuries: Medial head strain; contusion

Rehabilitation: Progressive loading; screen vascular red flags if acute calf swelling

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sartorius

ID: KNEE-MUS-10

Muscle: Sartorius

Origin: ASIS

Insertion: Pes anserinus

Innervation: Femoral nerve (L2-L3)

Blood Supply: Femoral artery branches

Fiber Direction: Oblique longest muscle

Primary Action: Hip FABER; knee flexion and tibial IR

Secondary Action: Pes medial support

Stabilizing Function: Minor dynamic medial stabilizer

Synergists: Gracilis, ST at pes

Antagonists: Glute max / vastii depending on action

Length-Tension Relationship: Active across hip and knee — multiplanar

EMG Activation: FABER motions, early swing

Trigger Points: Anterior thigh strip

Pain Referral: Anterior medial knee

Stretch: Combined extension-adduction-IR stretch carefully

Strengthening: FABER loading, step-ups

Neuromuscular Control: Pes anserine load management

Clinical Importance: Pes anserine bursitis differential; ASIS avulsion youth

Common Injuries: Proximal avulsion; strain

Rehabilitation: Progressive multiplanar loading

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Gracilis

ID: KNEE-MUS-11

Muscle: Gracilis

Origin: Inferior pubic ramus / body

Insertion: Pes anserinus

Innervation: Obturator nerve (L2-L3)

Blood Supply: Obturator artery

Fiber Direction: Medial thigh vertical

Primary Action: Hip adduction; knee flexion and tibial IR

Secondary Action: Pes medial stability

Stabilizing Function: Dynamic medial restraint assistant

Synergists: Other adductors, ST, sartorius

Antagonists: Abductors, VL

Length-Tension Relationship: Common ACL graft with ST (hamstring graft)

EMG Activation: Adductor and pes tasks

Trigger Points: Medial thigh

Pain Referral: Medial knee

Stretch: Butterfly / adductor stretch graded

Strengthening: Copenhagen, squeezes, curls

Neuromuscular Control: Graft harvest rehab after ACLR

Clinical Importance: Adductor-related groin pain; harvest morbidity

Common Injuries: Strain; harvest

Rehabilitation: Progressive adductor and knee flexion strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tensor Fasciae Latae

ID: KNEE-MUS-12

Muscle: Tensor fasciae latae

Origin: ASIS / anterior iliac crest

Insertion: ITB to Gerdy's tubercle

Innervation: Superior gluteal nerve (L4-S1)

Blood Supply: Superior gluteal artery

Fiber Direction: Anterolateral into ITB

Primary Action: Hip flexion, abduction, IR; tensions ITB

Secondary Action: Assists knee stability via ITB near extension

Stabilizing Function: Lateral knee restraint via ITB

Synergists: Glute med (abd), RF (flex)

Antagonists: Adductors

Length-Tension Relationship: Overactivity common when glute med weak

EMG Activation: Side-step and flexion-abduction tasks

Trigger Points: Lateral hip/thigh

Pain Referral: Lateral knee (ITBS continuum)

Stretch: Hip extension-adduction stretch; foam roll adjunct (evidence mixed)

Strengthening: Usually prioritize glute med over TFL dominance

Neuromuscular Control: Pelvic control in running

Clinical Importance: IT band syndrome contributor; PF lateral force

Common Injuries: Myofascial overuse

Rehabilitation: Cadence/load management; hip abductor endurance; running form

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

6. Tendons

RECORD: Patellar Tendon

ID: KNEE-TEN-01

Tendon: Patellar tendon (ligamentum patellae)

Associated Muscle: Quadriceps via patella

Insertion: Tibial tuberosity

Blood Supply: Inferior medial/lateral genicular; fat pad vessels — proximal posterior watershed relevant to tendinopathy

Biomechanics: Transmits extensor force; high load in jump-land (energy storage/release)

Healing: Tendinopathy is collagen disrepair continuum (Cook & Purdam) — responds to progressive loading over months

Clinical Assessment: Localized inferior pole pain; decline squat pain; load-related pattern; rule out Osgood in youth

Ultrasound Findings: Thickening, hypoechoic zones, neovascularization (Doppler) at proximal tendon common

MRI Findings: Increased T2 signal proximal tendon; rule out partial tear/fat pad

Common Injuries: Patellar tendinopathy (jumper's knee); rupture (rare — often systemic risk or prior injections)

Treatment: Education, load management, progressive tendon loading; surgery rare for tendinopathy

Exercises: Isometric hold (e.g. Spanish squat) for analgesia → heavy slow resistance → energy storage/plyometrics

Rehabilitation: Pain-monitoring model (e.g. tolerable during/after); 12+ weeks typical; address kinetic chain

Return to Sport: Criteria: pain-tolerable training loads, hop/jump capacity, sport volume built gradually

References: Cook JL & Purdam CR continuum. Malliaras P patellar tendinopathy. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Quadriceps Tendon

ID: KNEE-TEN-02

Tendon: Quadriceps tendon

Associated Muscle: Rectus femoris and vastii

Insertion: Patellar base / anterior patella

Blood Supply: Genicular anastomosis — ruptures more in older adults with degeneration

Biomechanics: Extensor force to patella; rupture loses active extension

Healing: Complete rupture needs surgical repair; tendinopathy managed with loading

Clinical Assessment: Suprapatellar pain; palpable gap if rupture; unable to SLR/extend actively if complete

Ultrasound Findings: Fiber disruption, hematoma; dynamic gap

MRI Findings: Confirms partial vs complete; retraction

Common Injuries: Tendinopathy; rupture (often >40 years, systemic risk factors)

Treatment: Complete rupture — urgent ortho repair; tendinopathy — progressive loading

Exercises: Isometrics → progressive closed chain → eccentrics as tolerated

Rehabilitation: Post-repair: protected ROM/brace per surgeon; delayed heavy loading

Return to Sport: Often 6-9+ months post-repair; tendinopathy criteria-based earlier

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Quad tendon rupture reviews.

RECORD: Hamstring Tendons

ID: KNEE-TEN-03

Tendon: Distal and proximal hamstring tendons (knee module focus distal BF/ST/SM + graft tendons)

Associated Muscle: Biceps femoris, semitendinosus, semimembranosus

Insertion: Fibular head (BF); pes anserinus (ST); posteromedial tibia (SM)

Blood Supply: Perforators; watershed zones near free tendon

Biomechanics: Eccentric late-swing sprint; distal BF lateral stability with PLC

Healing: Muscle-tendon strains remodel with progressive lengthening loads; proximal free-tendon slower

Clinical Assessment: Palpation map; strength; Nordic capacity; differentiate sciatic referred pain

Ultrasound Findings: Useful for distal/proximal tendon integrity; operator-dependent

MRI Findings: BAMIC grading for intramuscular; retraction for free-tendon tears

Common Injuries: LH strain; distal BF; ST/gracilis harvest morbidity after ACLR

Treatment: Progressive criteria-based rehab; surgery for selected proximal avulsions

Exercises: Isometrics → heavy slow → Nordics → sprint mechanics

Rehabilitation: Askling/BAMIC-informed; avoid rapid RTS

Return to Sport: Strength symmetry, H-test, sprint exposure progressive

References: Askling C. British Athletics BAMIC. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Popliteus Tendon

ID: KNEE-TEN-04

Tendon: Popliteus tendon

Associated Muscle: Popliteus

Insertion: Posterior tibia (muscle belly); tendon originates lateral femoral condyle

Blood Supply: Lateral genicular / popliteal branches

Biomechanics: Dynamic PLC; unlocks screw-home; resists tibial ER

Healing: With PLC injuries — often surgical if grade III complex

Clinical Assessment: Posterolateral pain; dial test; PLC cluster

Ultrasound Findings: Limited deep visualization

MRI Findings: Tendon signal/discontinuity; associated LCL/PFL

Common Injuries: PLC trauma; rare isolated tendinopathy

Treatment: Per PLC grade; protect rotational stress

Exercises: Tibial IR control; progressive rotational stability after clearance

Rehabilitation: PLC protocols dominate timeline

Return to Sport: Tied to PLC/multi-ligament RTS criteria

References: LaPrade PLC. Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pes Anserinus

ID: KNEE-TEN-05

Tendon: Pes anserinus complex (sartorius, gracilis, semitendinosus tendons)

Associated Muscle: Sartorius, gracilis, semitendinosus

Insertion: Anteromedial proximal tibia distal to medial joint line

Blood Supply: Inferior medial genicular / soft tissue

Biomechanics: Medial knee stabilization; resists valgus/ER dynamically

Healing: Bursitis/tendinopathy responds to load modification and progressive strength

Clinical Assessment: Tenderness below medial joint line; swelling; differentiate MCL/meniscus

Ultrasound Findings: Bursal fluid; tendon thickening

MRI Findings: Bursitis; rules out meniscal/MCL pathology

Common Injuries: Pes anserine bursitis; insertional irritation in cyclists/swimmers/OA patients

Treatment: Load management, flexibility, strengthening; injection selective

Exercises: Quad/hip strength; gradual hamstring-pes loading; address training errors

Rehabilitation: Reduce provocative adduction sitting loads; progressive return

Return to Sport: When tenderness and strength allow sport-specific volume

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

7. Bursae

RECORD: Suprapatellar Bursa

ID: KNEE-BUR-01

Bursa: Suprapatellar bursa/pouch

Location: Deep to quad tendon, proximal to patella — communicates with knee joint in adults

Structures Protected: Quadriceps tendon from femur

Function: Allows smooth extensor glide; reflects intra-articular effusion

Clinical Importance: Effusion visible/palpable here; communicates with joint

Common Disorders: Effusion from intra-articular pathology; bursitis rare as isolated

Clinical Examination: Stroke test, ballottement, bulge sign

Ultrasound: Excellent for effusion quantification

MRI: Fluid signal continuous with joint

Treatment: Treat cause of effusion; aspiration if indicated

Rehabilitation: Quad activation despite effusion; address intra-articular drivers

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Prepatellar Bursa

ID: KNEE-BUR-02

Bursa: Prepatellar bursa

Location: Subcutaneous anterior to patella

Structures Protected: Skin from patella

Function: Reduce friction during kneeling

Clinical Importance: Housemaid's knee — swelling localized anterior patella (not joint effusion)

Common Disorders: Traumatic/inflammatory bursitis; septic bursitis (urgent)

Clinical Examination: Discrete anterior swelling; knee ROM often preserved vs septic arthritis

Ultrasound: Fluid collection superficial to patella; guide aspiration

MRI: If deep infection or unclear diagnosis

Treatment: Protection, NSAIDs if appropriate; antibiotics/drainage if septic; avoid recurrent kneeling trauma

Rehabilitation: Padding, activity modification, gradual kneeling exposure

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Superficial Infrapatellar Bursa

ID: KNEE-BUR-03

Bursa: Superficial infrapatellar bursa

Location: Between skin and patellar tendon / tibial tuberosity

Structures Protected: Skin from patellar tendon

Function: Kneeling friction reduction

Clinical Importance: Clergyman's knee continuum with kneeling bursitis

Common Disorders: Bursitis from repetitive kneeling

Clinical Examination: Swelling over tibial tuberosity/tendon — differentiate Osgood in youth

Ultrasound: Superficial fluid

MRI: If needed to exclude tendon pathology

Treatment: Load modification, padding; rare injection

Rehabilitation: Kneeling hygiene; tendon loading if concurrent tendinopathy

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Deep Infrapatellar Bursa

ID: KNEE-BUR-04

Bursa: Deep infrapatellar bursa

Location: Between patellar tendon and tibia, below fat pad

Structures Protected: Patellar tendon from tibia

Function: Reduce tendon-bone friction

Clinical Importance: Can accompany fat pad / patellar tendinopathy pain maps

Common Disorders: Bursitis; often secondary

Clinical Examination: Deep pain near distal pole/tuberosity; less visible swelling

Ultrasound: Deep fluid relative to tendon

MRI: Fluid deep to tendon; assess fat pad

Treatment: Treat associated tendon/fat pad drivers

Rehabilitation: Patellar tendon progressive loading principles

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pes Anserine Bursa

ID: KNEE-BUR-05

Bursa: Pes anserine bursa

Location: Between pes tendons and MCL / medial tibia

Structures Protected: Pes tendons from MCL/tibia

Function: Reduce friction of pes complex

Clinical Importance: Medial knee pain below joint line — common in OA and overweight patients

Common Disorders: Pes anserine bursitis

Clinical Examination: Point tenderness; pain with resisted knee flexion; differentiate meniscus/MCL

Ultrasound: Bursal fluid

MRI: Confirms; excludes other medial pathology

Treatment: Load modification, stretching/strengthening, injection selective

Rehabilitation: Quad/hip strength; gradual return to provocative activity

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Semimembranosus Bursa (Baker Cyst Region)

ID: KNEE-BUR-06

Bursa: Semimembranosus-medial gastrocnemius bursa (related to Baker's cyst)

Location: Posteromedial knee between SM tendon and medial gastrocnemius head

Structures Protected: Tendons in popliteal region

Function: Reduce friction; may communicate with joint

Clinical Importance: Baker's (popliteal) cyst enlargement — often secondary to intra-articular disease

Common Disorders: Baker's cyst; rupture mimics DVT (must differentiate)

Clinical Examination: Popliteal mass; fullness in extension; screen for effusion/OA/meniscus

Ultrasound: First-line for cyst vs DVT differentiation support

MRI: Defines cyst and intra-articular drivers

Treatment: Treat underlying knee pathology; aspiration/injection selective; surgery rare

Rehabilitation: Quad strength, effusion control, address meniscal/OA drivers

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Baker cyst reviews.

8. Nerves

RECORD: Femoral Nerve

ID: KNEE-NRV-01

Nerve: Femoral nerve (quadriceps motor; saphenous sensory continuation)

Root Levels: L2-L4

Motor Function: Iliacus, sartorius, pectineus, quadriceps

Sensory Distribution: Anterior thigh; saphenous to medial leg/foot

Dermatomes: Primarily L3-L4 anterior thigh/medial leg via saphenous

Myotomes: L3-L4 knee extension

Entrapment Sites: Retroperitoneal/iliacus hematoma; under inguinal ligament; iatrogenic

Clinical Presentation: Quad weakness, reduced patellar reflex, anterior thigh sensory loss

Neurodynamic Tests: Femoral nerve stretch (prone knee bend) — interpret cautiously

Differential Diagnosis: L3-L4 radiculopathy; AMI from knee effusion (not true neuropathy)

EMG: Confirms femoral neuropathy vs root vs AMI patterns

Imaging: MRI lumbar/plexus if indicated

Treatment: Address cause; protect falls from quad weakness

Rehabilitation: Quad re-education, NMES, gait aid, progressive strengthening

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Saphenous Nerve

ID: KNEE-NRV-02

Nerve: Saphenous nerve (including infrapatellar branch)

Root Levels: L3-L4 (femoral continuation)

Motor Function: None (pure sensory)

Sensory Distribution: Medial knee (infrapatellar branch), medial leg to medial foot

Dermatomes: Medial leg L4 typically

Myotomes: N/A

Entrapment Sites: Hunter's (adductor) canal; infrapatellar branch neuroma after medial knee surgery/trauma

Clinical Presentation: Medial knee burning/numbness; pain with kneeling; Tinel at scar

Neurodynamic Tests: Saphenous neurodynamics / slump variants

Differential Diagnosis: MCL, pes, medial meniscus, L4 radiculopathy

EMG: Sensory NCS may help; often clinical diagnosis for infrapatellar neuroma

Imaging: US/MRI for neuroma in selected cases

Treatment: Desensitization, neurodynamics, topical agents, injection/neuroma procedures selected

Rehabilitation: Scar mobility, graded exposure, strength around knee

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Infrapatellar saphenous neuroma literature. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Tibial Nerve (at Knee)

ID: KNEE-NRV-03

Nerve: Tibial nerve in popliteal fossa

Root Levels: L4-S3

Motor Function: Hamstrings (tibial portions), posterior compartment distal, intrinsics via plantar nerves

Sensory Distribution: Posterolateral leg via sural contributions; plantar foot via medial/lateral plantar

Dermatomes: S1-S2 plantar emphasis

Myotomes: S1-S2 plantarflexion / toe flexion

Entrapment Sites: Popliteal fossa mass/cyst compression; more distal tarsal tunnel

Clinical Presentation: Posterior pain/paresthesia; intrinsic weakness if distal

Neurodynamic Tests: SLR with tibial bias (dorsiflexion/eversion)

Differential Diagnosis: Baker cyst, DVT, S1 radiculopathy, PCL injury pain

EMG: Localizes tibial neuropathy

Imaging: US/MRI popliteal fossa

Treatment: Treat compressive cause; neurodynamics

Rehabilitation: Nerve mobility, address cyst drivers, strength

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Common Fibular (Peroneal) Nerve

ID: KNEE-NRV-04

Nerve: Common fibular nerve at fibular neck

Root Levels: L4-S2

Motor Function: Short head BF; then deep/superficial fibular — dorsiflexors/evertors

Sensory Distribution: Lateral leg and dorsum foot

Dermatomes: L5 dorsal foot emphasis

Myotomes: L4-L5 dorsiflexion/eversion

Entrapment Sites: Fibular neck — cast, habit leg crossing, trauma, PLC surgery risk

Clinical Presentation: Foot drop, steppage gait, sensory loss dorsum foot

Neurodynamic Tests: SLR with fibular bias (plantarflexion/inversion)

Differential Diagnosis: L4-L5 radiculopathy, compartment syndrome, sciatic injury

EMG: Critical for localization and prognosis

Imaging: US at fibular neck; MRI if mass/trauma

Treatment: Relieve compression, AFO as needed, surgical decompression selected

Rehabilitation: AFO gait, dorsiflexion strengthening/NMES, skin protection

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Obturator Nerve

ID: KNEE-NRV-05

Nerve: Obturator nerve (gracilis / adductor relevance to medial knee)

Root Levels: L2-L4

Motor Function: Adductors, gracilis, obturator externus

Sensory Distribution: Small medial thigh cutaneous field (variable to knee)

Dermatomes: Medial thigh L2-L3

Myotomes: L2-L3 hip adduction

Entrapment Sites: Obturator canal; pelvic surgery; less commonly at knee

Clinical Presentation: Medial thigh pain/weak adduction; may refer toward medial knee

Neurodynamic Tests: Hip extension-abduction biased neurodynamics

Differential Diagnosis: Adductor strain, athletic pubalgia, L2-L3 radiculopathy, pes pain

EMG: Confirms obturator neuropathy

Imaging: MRI pelvis/thigh if indicated

Treatment: Address entrapment cause; load management adductors

Rehabilitation: Adductor progressive loading; differential motor control

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

9. Blood Supply

RECORD: Popliteal Artery

ID: KNEE-ART-01

Artery: Popliteal artery

Origin: Continuation of femoral artery after hiatus adductorius

Course: Through popliteal fossa deep to tibial nerve/vein orientation variations; divides at lower border soleus into anterior and posterior tibial

Branches: Genicular arteries (SM/SL/IM/IL/middle); sural arteries

Structures Supplied: Knee joint via genicular anastomosis; posterior capsule; surrounding muscles

Venous Drainage: Popliteal vein → femoral vein

Clinical Importance: Dislocation of knee — limb-threatening vascular injury risk; always neurovascular exam; ABI/CTA if concern

Imaging: Duplex, CTA, angiography for trauma

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. Knee dislocation vascular injury guidelines.

RECORD: Superior Medial Genicular

ID: KNEE-ART-02

Artery: Superior medial genicular artery

Origin: Popliteal artery

Course: Winds above medial femoral condyle deep to muscles toward anterior anastomosis

Branches: Articular and soft-tissue rami

Structures Supplied: Medial femoral condyle, vastus medialis, medial capsule, ACL region anastomoses

Venous Drainage: Genicular veins to popliteal

Clinical Importance: Part of genicular anastomosis; target in genicular artery embolization for knee OA pain (emerging)

Imaging: Angiography/CTA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Superior Lateral Genicular

ID: KNEE-ART-03

Artery: Superior lateral genicular artery

Origin: Popliteal artery

Course: Above lateral femoral condyle to anterior anastomosis

Branches: Articular rami

Structures Supplied: Lateral femoral condyle, VL, lateral capsule, PF anastomosis

Venous Drainage: Genicular veins to popliteal

Clinical Importance: Anastomosis redundancy if one genicular injured; surgical approach awareness

Imaging: Angiography/CTA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Inferior Medial Genicular

ID: KNEE-ART-04

Artery: Inferior medial genicular artery

Origin: Popliteal artery

Course: Below medial tibial plateau along MCL region forward

Branches: Articular rami

Structures Supplied: Medial tibial plateau, MCL region, inferior PF anastomosis, patellar tendon region

Venous Drainage: Genicular veins to popliteal

Clinical Importance: Patellar blood supply contribution; medial surgery awareness

Imaging: Angiography/CTA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Inferior Lateral Genicular

ID: KNEE-ART-05

Artery: Inferior lateral genicular artery

Origin: Popliteal artery

Course: Passes above fibular head / near LCL toward anterior anastomosis

Branches: Articular rami

Structures Supplied: Lateral tibial plateau, LCL region, inferior PF anastomosis

Venous Drainage: Genicular veins to popliteal

Clinical Importance: PLC surgical proximity; anastomosis participant

Imaging: Angiography/CTA

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH.

RECORD: Middle Genicular

ID: KNEE-ART-06

Artery: Middle genicular artery

Origin: Popliteal artery

Course: Pierces oblique popliteal ligament into intercondylar notch

Branches: Branches to cruciate ligaments and synovial notch tissues

Structures Supplied: ACL and PCL primary blood supply; synovium of notch

Venous Drainage: To popliteal venous system

Clinical Importance: Cruciate healing/vascularity; notch surgery awareness

Imaging: Angiography (not routine clinically)

References: Standring S. Gray's Anatomy. Moore KL et al. Clinically Oriented Anatomy. Netter FH. ACL vascular anatomy literature.

10. Biomechanics

RECORD: Walking

ID: KNEE-BIO-01

Movement: Walking gait — knee

Plane: Sagittal primary

Axis: Transepicondylar approximate

Primary Movers: Quadriceps early stance; hamstrings late swing; gastroc/soleus

Secondary Movers: Popliteus unlock; hip abductors for alignment

Stabilizers: ACL/PCL/collaterals passive; menisci load share

Arthrokinematics: Screw-home into extension; flexion with femoral roll-glide pattern

Osteokinematics: Stance flexion ~15-20 deg; swing flexion ~60 deg typical

Joint Reaction Forces: Often about 2-3x BW peak (model-dependent)

Torque: Extensor then flexor moments phasic

Length-Tension: Quad works near mid-length early stance

EMG: Phasic low-moderate vs running

Functional Activities: Community ambulation

Sport Applications: Base for RTS progressions

Compensations: Stiff-knee gait; vaulting; antalgic shortened stance

Common Dysfunctions: OA limited extension; quad weakness

Clinical Assessment: Observational gait; ROM; strength

Corrective Exercises: Gait drill, quad strength, extension mobility

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Running

ID: KNEE-BIO-02

Movement: Running

Plane: Multiplanar

Axis: Multi-joint

Primary Movers: Quads absorb; glutes/hamstrings propulsion coupling

Secondary Movers: Gastroc; hip abductors

Stabilizers: ACL under rotational/valgus loads if poor mechanics

Arthrokinematics: Greater flexion excursions than walking; rapid transitions

Osteokinematics: Flight + stance; cadence/stride tradeoff

Joint Reaction Forces: Often 4-8x BW estimates vary by speed/style

Torque: High extensor demand early stance

Length-Tension: Tendon energy storage important

EMG: High burst EMG

Functional Activities: Fitness/field running

Sport Applications: All running sports

Compensations: Overstride; dynamic valgus; crossover gait

Common Dysfunctions: PFPS, ITBS, tendinopathy, bone stress

Clinical Assessment: Video gait; step rate; pain map

Corrective Exercises: Cadence cues, run-walk, hip/quad capacity

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Sprinting

ID: KNEE-BIO-03

Movement: Sprinting

Plane: Sagittal dominant

Axis: Multi-joint

Primary Movers: Hamstrings late swing; iliopsoas/RF swing; glute propulsion

Secondary Movers: Gastroc push-off

Stabilizers: ACL risk on plant/cut more than pure straight sprint; hamstring high risk

Arthrokinematics: Rapid flexion-extension; extreme angular velocities

Osteokinematics: Max velocity mechanics

Joint Reaction Forces: Very high soft-tissue loads

Torque: Huge hamstring eccentric torque late swing

Length-Tension: Lengthening hamstrings vulnerable

EMG: Max EMG hamstring late swing

Functional Activities: Track sprinting

Sport Applications: Athletics, field sport speed

Compensations: Early overstride; anterior pelvic tilt excess

Common Dysfunctions: Hamstring strain

Clinical Assessment: Sprint mechanics video; Nordic capacity

Corrective Exercises: Askling drills, progressive exposures

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Jumping

ID: KNEE-BIO-04

Movement: Jump takeoff

Plane: Sagittal

Axis: Multi-joint

Primary Movers: Quads, glutes, plantarflexors triple extension

Secondary Movers: Hamstring co-contraction

Stabilizers: Extensor mechanism / patellar tendon high load

Arthrokinematics: Rapid extension from countermovement flexion

Osteokinematics: CMJ pattern

Joint Reaction Forces: Multiple x BW

Torque: High knee extensor torque

Length-Tension: Tendon spring contribution

EMG: Explosive quad/glute EMG

Functional Activities: Jump sports

Sport Applications: Basketball/volleyball

Compensations: Knee-only strategy without hip

Common Dysfunctions: Tendinopathy risk if volume spike

Clinical Assessment: CMJ testing

Corrective Exercises: Strength base then plyometric ladder

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Landing

ID: KNEE-BIO-05

Movement: Jump landing

Plane: Multiplanar

Axis: Multi-joint

Primary Movers: Eccentric quads/glutes

Secondary Movers: Hamstrings; hip ERs/abductors

Stabilizers: ACL vulnerable with valgus + IR + shallow flexion

Arthrokinematics: Rapid flexion with controlled rotation

Osteokinematics: Soft vs stiff landing strategies

Joint Reaction Forces: Very high transient loads

Torque: Frontal plane moments linked to ACL risk

Length-Tension: Quad eccentric dominant

EMG: High eccentric EMG

Functional Activities: Drop land, cut decelerations

Sport Applications: All jump-land sports

Compensations: Stiff land; dynamic valgus; hip drop

Common Dysfunctions: ACL, PFPS

Clinical Assessment: Drop-jump assessment

Corrective Exercises: Soft-land cues, hip strategy, single-leg progressions

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Squat

ID: KNEE-BIO-06

Movement: Squat

Plane: Sagittal dominant

Axis: Multi-joint

Primary Movers: Quads and glutes

Secondary Movers: Adductors deep; erectors trunk

Stabilizers: Menisci/ACL depending on depth/valgus

Arthrokinematics: Tibiofemoral roll-glide with depth; PF contact rises with flexion

Osteokinematics: Coupled hip-knee-ankle flexion

Joint Reaction Forces: PF and TF forces rise with depth/load

Torque: Knee extensor moment increases with forward knee travel (style-dependent)

Length-Tension: Quad lengthens under load

EMG: High VL/VM EMG

Functional Activities: ADLs sit-to-stand

Sport Applications: Strength training

Compensations: Knee valgus; heel rise; lumbar flexion

Common Dysfunctions: PF pain deep squat; meniscal symptoms

Clinical Assessment: Bodyweight squat screen

Corrective Exercises: Box squat to tolerable depth; tempo; hip hinge mix

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Lunge

ID: KNEE-BIO-07

Movement: Lunge

Plane: Sagittal + frontal control

Axis: Multi-joint

Primary Movers: Lead-leg quad/glute

Secondary Movers: Trail hip flexor; adductors

Stabilizers: Frontal knee alignment critical

Arthrokinematics: Lead knee flexion with tibial translation control

Osteokinematics: Split-stance osteokinematics

Joint Reaction Forces: High PF demand on lead limb

Torque: Unilateral extensor torque

Length-Tension: Similar to single-leg squat demands

EMG: Unilateral EMG high

Functional Activities: Stairs, running mechanics prep

Sport Applications: Field sport strength

Compensations: Lead knee valgus; trunk lean

Common Dysfunctions: PFPS, patellar tendinopathy if overloaded

Clinical Assessment: Forward/reverse lunge quality

Corrective Exercises: Short-range → full; trunk upright cues

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Cutting

ID: KNEE-BIO-08

Movement: Cutting / COD

Plane: Multiplanar

Axis: Multi-joint

Primary Movers: Quads decelerate; glute med/max; adductors

Secondary Movers: Hamstrings; trunk

Stabilizers: ACL and MCL high demand with valgus/rotation

Arthrokinematics: Rapid plant-limb flexion with tibial rotation

Osteokinematics: Direction change

Joint Reaction Forces: High combined loads

Torque: Large frontal/transverse moments

Length-Tension: Plant limb at vulnerable lengths

EMG: High co-contraction EMG

Functional Activities: Evasion

Sport Applications: Football, soccer, basketball

Compensations: Lateral trunk lean; stiff plant; knee collapse

Common Dysfunctions: ACL injury mechanism

Clinical Assessment: COD tests; 2D frontal projection

Corrective Exercises: Technique + strength + plyometrics + graded COD

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Pivoting

ID: KNEE-BIO-09

Movement: Pivoting

Plane: Transverse dominant

Axis: Longitudinal tibial/femoral

Primary Movers: Hamstrings rotational control; popliteus; quads

Secondary Movers: Hip rotators

Stabilizers: ACL primary rotational stabilizer

Arthrokinematics: Tibial IR/ER on femur with flexed knee

Osteokinematics: Pivot on planted foot

Joint Reaction Forces: High shear/rotation

Torque: Rotational torque through knee

Length-Tension: Depends on flexion angle

EMG: PLC/ACL related activation

Functional Activities: Court sports turns

Sport Applications: Soccer, skiing

Compensations: Kneeing-in; spinning on locked knee

Common Dysfunctions: ACL tear, meniscal tear

Clinical Assessment: Pivot shift clinically (laxity); movement quality

Corrective Exercises: Rotational control progressions after ACLR

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Stair Climbing

ID: KNEE-BIO-10

Movement: Stair ascent/descent

Plane: Sagittal

Axis: Multi-joint

Primary Movers: Ascent: concentric quad/glute. Descent: eccentric quad

Secondary Movers: Hip abductors

Stabilizers: High PF joint stress especially descent

Arthrokinematics: Greater knee flexion than level walking

Osteokinematics: Step-over-step patterns

Joint Reaction Forces: PF forces often high on descent (study-dependent multiples of BW)

Torque: Large eccentric knee extensor torque descending

Length-Tension: Quad eccentrically loaded

EMG: High VM/VL EMG

Functional Activities: ADLs

Sport Applications: Stadium training

Compensations: Step-to pattern; handrail lean; leading with same leg

Common Dysfunctions: PFPS, OA pain

Clinical Assessment: Step-down test; pain mapping

Corrective Exercises: Step-ups/downs progressive; hip strength; cadence

References: Neumann DA. Kinesiology of the Musculoskeletal System. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

11. Pathologies

RECORD: ACL Tear

ID: KNEE-PATH-01

Condition: ACL tear

Definition: Partial or complete disruption of ACL fibers with anterior-rotational laxity

Mechanism: Noncontact pivot (deceleration, cut, land) most common; contact valgus also

Risk Factors: Female sex (sport-specific), prior ACL, cutting sports, neuromuscular factors, anatomy (notch, slope) debated contributors

Symptoms: Pop, rapid effusion, giving-way, loss of confidence in cutting

Red Flags: Locked knee (bucket-handle), fracture, neurovascular injury, multi-ligament/dislocation

Differential Diagnosis: MCL, meniscus, patellar dislocation, PCL

Clinical Examination: Effusion, ROM, Lachman, pivot, associated structures

Special Tests: Lachman, pivot shift, anterior drawer

Imaging: MRI confirms; radiographs for Segond/fracture

Healing Timeline: Does not reliably heal complete midsubstance tears; graft maturation many months after ACLR

Conservative Management: Coper algorithm, bracing, progressive NMC rehab for selected patients

Surgical Management: ACLR (autograft BTB/hamstring/quad) for instability/sport; repair selected proximal tears evolving evidence

Rehabilitation Phases: Early ROM/quad → strength → plyometrics → COD → RTS testing

Return to Running: Often after 3-4+ months if criteria met post-ACLR — individualize

Return to Sport: Commonly 9-12+ months with passing criteria; reinjury risk if early

Prognosis: Good with appropriate surgery/rehab; osteoarthritis risk long-term still elevated

Clinical Guidelines: AAOS/AOSSM-aligned shared decision themes; criteria-based RTS literature

Evidence Level: Strong evidence for Lachman/MRI diagnosis; RTS criteria reduce but do not eliminate reinjury

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: PCL Tear

ID: KNEE-PATH-02

Condition: PCL tear

Definition: Disruption of PCL with posterior tibial laxity

Mechanism: Dashboard; fall on flexed knee plantarflexed foot

Risk Factors: High-energy trauma; combined PLC

Symptoms: Vague posterior pain, difficulty deceleration, posterior sag

Red Flags: Multi-ligament, vascular injury if dislocation history

Differential Diagnosis: PLC, ACL, posteromedial pain generators

Clinical Examination: Posterior drawer/sag, quad active test, dial

Special Tests: Posterior drawer, sag, dial

Imaging: MRI; stress radiography selected

Healing Timeline: Many isolated grade I-II improve over 6-12 weeks protected rehab

Conservative Management: Extension bracing early, quad focus, avoid early isolated hamstring shear

Surgical Management: Grade III/combined/bony avulsion may need surgery

Rehabilitation Phases: Protect posterior sag → progressive strength → functional

Return to Running: When effusion/strength/control allow

Return to Sport: Criteria-based; longer if reconstructed

Prognosis: Generally favorable for isolated treated appropriately

Clinical Guidelines: PCL management reviews

Evidence Level: Moderate — many recommendations consensus-based

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: MCL Injury

ID: KNEE-PATH-03

Condition: MCL injury

Definition: Sprain/tear of superficial ± deep MCL

Mechanism: Valgus force; cutting

Risk Factors: Contact sports; ACL concomitant

Symptoms: Medial pain, valgus stress pain, mild-moderate laxity

Red Flags: Multi-ligament; fracture; locked knee

Differential Diagnosis: Medial meniscus, pes bursitis, medial tibial plateau fracture

Clinical Examination: Valgus stress 0/30, joint line exam

Special Tests: Valgus stress test

Imaging: MRI if high grade/unsure/concomitant; US adjunct

Healing Timeline: Grade I-II often 1-6 weeks; III longer

Conservative Management: Hinged brace, early ROM, progressive loading — excellent healing

Surgical Management: Rare except Stener-like distal / multi-ligament

Rehabilitation Phases: Protect valgus early → strength → COD

Return to Running: When pain/laxity allow jog progression

Return to Sport: Grade-dependent weeks to months

Prognosis: Excellent for isolated

Clinical Guidelines: Standard sports medicine MCL pathways

Evidence Level: Good clinical consensus evidence for nonop grade I-II

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: LCL Injury

ID: KNEE-PATH-04

Condition: LCL injury

Definition: Sprain/tear of LCL often with PLC

Mechanism: Varus trauma

Risk Factors: High-energy; thin LCL anatomy

Symptoms: Lateral pain, varus laxity

Red Flags: PLC, common fibular nerve, vascular if high energy

Differential Diagnosis: ITBS, meniscus, PTFJ

Clinical Examination: Varus stress, dial, nerve exam

Special Tests: Varus stress, dial, posterolateral drawer

Imaging: MRI; radiographs for arcuate sign

Healing Timeline: Low grade weeks; grade III often surgical timeline months

Conservative Management: Brace protection for low grade

Surgical Management: Often reconstruct if grade III/PLC

Rehabilitation Phases: Protect varus → rotational control → RTS

Return to Running: Delayed until stability restored

Return to Sport: Months if surgical

Prognosis: Depends on PLC recognition

Clinical Guidelines: LaPrade-aligned PLC approach

Evidence Level: Moderate evidence; anatomy-driven

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Meniscal Tear

ID: KNEE-PATH-05

Condition: Meniscal tear

Definition: Disruption of medial or lateral meniscal fibrocartilage

Mechanism: Twist on flexed planted foot; degenerative cleavage

Risk Factors: ACL injury, age, squatting sports, discoid lateral

Symptoms: Joint-line pain, clicking, locking, effusion intermittent

Red Flags: True locking with loss of extension — urgent ortho; infection signs

Differential Diagnosis: PFPS, plica, OCD, loose body

Clinical Examination: Joint-line tenderness, Thessaly, McMurray

Special Tests: McMurray, Thessaly, Apley

Imaging: MRI; radiographs for OA

Healing Timeline: Vascular zone dependent; degenerative may stabilize symptomatically

Conservative Management: Exercise therapy first-line many degenerative tears (trial evidence)

Surgical Management: Repair if feasible; partial meniscectomy if irreparable; avoid unnecessary meniscectomy in degenerative OA

Rehabilitation Phases: Repair-protected vs meniscectomy accelerated pathways

Return to Running: Earlier after meniscectomy; delayed after repair

Return to Sport: Criteria + surgery-specific

Prognosis: Better if repaired when appropriate; OA risk after meniscectomy

Clinical Guidelines: ESSKA/AAOS-aligned meniscus preservation themes

Evidence Level: Stronger evidence favoring conservative care for many degenerative tears

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Patellofemoral Pain Syndrome

ID: KNEE-PATH-06

Condition: Patellofemoral pain syndrome (PFPS)

Definition: Anterior knee pain related to PF joint without other specific pathology explaining symptoms

Mechanism: Load exceeding capacity; training errors; movement factors multifactorial

Risk Factors: Adolescents/young adults, females, runners, weak hip/quad, abrupt load spikes

Symptoms: Diffuse anterior pain with stairs, squats, prolonged sitting (theater sign)

Red Flags: True locking, significant effusion, night pain systemic — reassess diagnosis

Differential Diagnosis: Tendinopathy, fat pad, plica, OA, referred hip

Clinical Examination: Step-down, squat, palpation, flexibility, hip strength

Special Tests: Limited specific tests; functional pain reproduction more useful

Imaging: Usually clinical; imaging to exclude other pathology

Healing Timeline: Symptoms often improve over weeks–months with active care

Conservative Management: Education, load management, hip + quad exercise (best evidence)

Surgical Management: Rare — only for specific structural problems not PFPS per se

Rehabilitation Phases: Symptom-guided progressive loading + movement retraining

Return to Running: Graded return with pain rules

Return to Sport: When training loads tolerated

Prognosis: Good with active rehab; recurrent if load mismanaged

Clinical Guidelines: PFPS consensus statements (e.g. British Journal of Sports Medicine consensus themes)

Evidence Level: Strong evidence for exercise therapy

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Patellar Tendinopathy

ID: KNEE-PATH-07

Condition: Patellar tendinopathy

Definition: Overuse insertional/proximal patellar tendon pain and dysfunction (continuum model)

Mechanism: Energy-storage overload (jumping); sudden volume increase

Risk Factors: Jump sports, males historically higher in some cohorts, limited ankle DF, high BMI debated

Symptoms: Localized inferior pole pain, load-related, morning stiffness mild

Red Flags: Rupture signs (rare); apophyseal pain in youth = Osgood not adult tendinopathy

Differential Diagnosis: PFPS, fat pad, Osgood (youth), SLJ

Clinical Examination: Decline squat pain, palpation proximal tendon

Special Tests: Royal London / decline squat clinical

Imaging: US/MRI if unclear or refractory

Healing Timeline: Collagen remodeling months — not 2-week fix

Conservative Management: Progressive tendon loading (isometrics → HSR → plyometrics); education

Surgical Management: Rare for pure tendinopathy

Rehabilitation Phases: Pain-monitoring progressive loading phases

Return to Running: When energy-storage drills tolerated

Return to Sport: Gradual return to jump volume

Prognosis: Good but often prolonged

Clinical Guidelines: Cook & Purdam / Malliaras loading frameworks

Evidence Level: Good evidence for progressive loading over passive-only care

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Quadriceps Tendinopathy

ID: KNEE-PATH-08

Condition: Quadriceps tendinopathy

Definition: Pain/dysfunction of quadriceps tendon typically at patellar base

Mechanism: Overload of extensor mechanism

Risk Factors: Jumping/lifting sports; age-related degeneration

Symptoms: Superior pole/patellar base pain with loading

Red Flags: Extensor lag / palpable gap suggests rupture

Differential Diagnosis: PFPS, bursitis, partial tear

Clinical Examination: Palpation, resisted extension pain

Special Tests: Functional loading tests

Imaging: US/MRI

Healing Timeline: Months with progressive loading

Conservative Management: Load management + progressive strengthening

Surgical Management: If rupture — repair

Rehabilitation Phases: Similar tendon continuum phases

Return to Running: When pain allows impact

Return to Sport: Criteria-based

Prognosis: Favorable if not ruptured

Clinical Guidelines: Tendinopathy management principles

Evidence Level: Moderate evidence extrapolated from patellar tendon literature

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Osgood-Schlatter Disease

ID: KNEE-PATH-09

Condition: Osgood-Schlatter disease

Definition: Traction apophysitis of tibial tuberosity in growing athletes

Mechanism: Repetitive quad traction on open tibial tubercle apophysis

Risk Factors: Ages ~10-15, sports with running/jumping, growth spurts

Symptoms: Activity-related tuberosity pain/swelling; prominence

Red Flags: Fever/redness infection; night pain tumor rare but consider if atypical

Differential Diagnosis: Patellar tendinopathy, SLJ, fracture, infection

Clinical Examination: Tenderness over tibial tuberosity; pain with resisted extension

Special Tests: Clinical diagnosis primarily

Imaging: Radiographs if atypical; avoid over-imaging

Healing Timeline: Self-limited with skeletal maturity months–years; symptoms wax/wane

Conservative Management: Load modification, ice, quad/hip strength, flexibility; rarely cast

Surgical Management: Rare — only complications/ossicle refractory selected

Rehabilitation Phases: Symptom-guided activity; maintain fitness cross-train

Return to Running: When symptoms allow — not forced through severe pain

Return to Sport: Usually before skeletal maturity ends symptoms

Prognosis: Excellent long-term

Clinical Guidelines: Pediatric sports medicine standard care

Evidence Level: Good observational/clinical consensus evidence

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Sinding-Larsen-Johansson Syndrome

ID: KNEE-PATH-10

Condition: Sinding-Larsen-Johansson syndrome

Definition: Traction apophysitis of inferior patellar pole in youth

Mechanism: Repetitive traction at proximal patellar tendon attachment on immature pole

Risk Factors: Young jumping athletes

Symptoms: Inferior pole pain with loading

Red Flags: Atypical systemic signs

Differential Diagnosis: Patellar tendinopathy, Osgood, fracture

Clinical Examination: Point tenderness inferior pole

Special Tests: Clinical

Imaging: Radiograph/MRI if needed

Healing Timeline: Self-limited with growth/load management

Conservative Management: Activity modification and progressive loading

Surgical Management: Rare

Rehabilitation Phases: Symptom-guided

Return to Running: Graded

Return to Sport: When tolerated

Prognosis: Excellent

Clinical Guidelines: Pediatric overuse guidelines

Evidence Level: Consensus clinical evidence

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: IT Band Syndrome

ID: KNEE-PATH-11

Condition: Iliotibial band syndrome

Definition: Lateral knee pain related to ITB–lateral femoral epicondyle region (impingement/compression paradigm evolving)

Mechanism: Repetitive knee flexion-extension around 20-30 deg (running)

Risk Factors: Runners, sudden volume increase, hip abductor weakness, cadence/stride factors

Symptoms: Sharp lateral pain that worsens with distance; often at same distance

Red Flags: Locked knee, significant swelling — reconsider

Differential Diagnosis: LCL, lateral meniscus, PF pain referral

Clinical Examination: Noble compression, Ober (limited specificity), single-leg stance pelvic control

Special Tests: Noble test classic but interpret cautiously

Imaging: Clinical primarily; MRI excludes other

Healing Timeline: Weeks with load modification typically

Conservative Management: Reduce running load, cadence increase, hip abductor endurance, progressive return

Surgical Management: Not indicated for primary ITBS

Rehabilitation Phases: Load → capacity → graded running

Return to Running: Run-walk then continuous

Return to Sport: When training distances pain-tolerable

Prognosis: Good

Clinical Guidelines: Running injury management reviews

Evidence Level: Moderate evidence for load and hip strengthening

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Baker's Cyst

ID: KNEE-PATH-12

Condition: Baker's (popliteal) cyst

Definition: Fluid-filled enlargement in popliteal bursa often communicating with knee joint

Mechanism: Secondary to effusion from OA, meniscal tear, inflammatory arthritis

Risk Factors: OA, meniscal pathology, inflammatory disease

Symptoms: Popliteal swelling/fullness; stiffness; rupture causes calf pain

Red Flags: DVT must be excluded if acute calf swelling/pain; infection

Differential Diagnosis: DVT, soft-tissue tumor (rare), gastroc strain

Clinical Examination: Popliteal mass; knee effusion exam

Special Tests: Clinical + US

Imaging: US first; MRI for intra-articular drivers

Healing Timeline: Tracks underlying disease; may persist

Conservative Management: Treat intra-articular cause; aspiration/CSI selective

Surgical Management: Rare cyst excision; address meniscus/OA as indicated

Rehabilitation Phases: Effusion control and strength for underlying knee

Return to Running: Per underlying condition

Return to Sport: Per underlying condition

Prognosis: Depends on driver

Clinical Guidelines: Standard ortho pathways

Evidence Level: Good diagnostic evidence for US vs DVT workflow

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Osteoarthritis

ID: KNEE-PATH-13

Condition: Knee osteoarthritis

Definition: Degenerative joint disease of TF and/or PF compartments

Mechanism: Cumulative load, injury legacy (ACL/meniscus), age, alignment, obesity

Risk Factors: Age, prior injury, obesity, female sex (some cohorts), genetics

Symptoms: Activity pain, stiffness <30-60 min, crepitus, swelling, reduced function

Red Flags: Night pain severe, fever, rapid hot swelling — infection/crystal/tumor workup

Differential Diagnosis: Meniscal symptoms, referred hip, inflammatory arthritis

Clinical Examination: Alignment, ROM, crepitus, effusion, function scores

Special Tests: No single special test — clinical + imaging

Imaging: Weight-bearing radiographs (Kellgren-Lawrence); MRI if atypical

Healing Timeline: Chronic progressive; flares weeks

Conservative Management: Exercise, weight loss, education first-line (OARSI/NICE themes); injections selective

Surgical Management: Osteotomy selected; arthroplasty when indicated

Rehabilitation Phases: Flare control → progressive strength → activity pacing

Return to Running: As tolerated with symptom rules

Return to Sport: Activity goals not always competitive sport

Prognosis: Variable; function improvable without reversing structure

Clinical Guidelines: OARSI / NICE osteoarthritis guidelines themes

Evidence Level: Strong evidence for exercise and weight loss

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Plica Syndrome

ID: KNEE-PATH-14

Condition: Plica syndrome

Definition: Symptomatic synovial plica (often mediopatellar) causing pain/clicking

Mechanism: Inflamed plica snapping over medial femoral condyle

Risk Factors: Trauma, overuse, immature PF tracking issues

Symptoms: Medial pain, clicking, theater sign overlap with PFPS

Red Flags: Locking true mechanical — meniscus/loose body

Differential Diagnosis: PFPS, medial meniscus, fat pad

Clinical Examination: Medial parapatellar tenderness; stutter test limited value

Special Tests: Clinical diagnosis of exclusion often

Imaging: MRI may show plica but poor specificity

Healing Timeline: Weeks–months

Conservative Management: Load modification, flexibility, quad/hip exercise, injection rare

Surgical Management: Arthroscopic resection if refractory after failed conservative

Rehabilitation Phases: As PFPS-like progressive loading

Return to Running: Graded

Return to Sport: When symptoms settle

Prognosis: Good if correctly selected surgical cases after failed care

Clinical Guidelines: Clinical reviews

Evidence Level: Limited high-level evidence; diagnosis challenging

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

RECORD: Osteochondritis Dissecans

ID: KNEE-PATH-15

Condition: Osteochondritis dissecans (OCD)

Definition: Idiopathic subchondral bone lesion with possible fragment instability — classic medial femoral condyle in youth

Mechanism: Multifactorial — ischemia/repetitive stress theories

Risk Factors: Juvenile athletes, males historically, repetitive impact

Symptoms: Activity pain, swelling, catching if loose fragment

Red Flags: Locked knee from loose body; infection signs

Differential Diagnosis: Meniscal tear, normal growing pain, AVN adult

Clinical Examination: Wilson test historically limited; effusion; mechanical symptoms

Special Tests: Clinical suspicion + imaging

Imaging: Radiographs (notch view); MRI staging stability

Healing Timeline: Juvenile open physis better healing; adult poorer

Conservative Management: Activity modification, unloading for stable juvenile lesions

Surgical Management: Fixation/drilling/restore cartilage for unstable lesions

Rehabilitation Phases: Protect WB per stage → progressive impact

Return to Running: Delayed until healing evidence

Return to Sport: Delayed; criteria + imaging

Prognosis: Better in juvenile stable lesions

Clinical Guidelines: Pediatric ortho OCD pathways

Evidence Level: Moderate evidence; staging-guided

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Specialty consensus literature as noted in guidelines field.

12. Rehabilitation Pathways

RECORD: ACL Reconstruction — Early Phase

ID: KNEE-REHAB-01

Condition: ACL reconstruction

Rehabilitation Phase: Early (approx 0-6 weeks — protocol-dependent)

Goals: Protect graft, full extension, quad activation, control effusion, gait with crutches as needed

Pain Management: Ice, elevation, analgesics per physician, effusion control

Weight Bearing Status: Usually WBAT with crutches unless concomitant meniscal repair/osteotomy dictates otherwise

ROM Goals: Full extension ASAP; flexion progressive to 90+ per protocol

Strength Goals: Quad sets, SLR without lag, NMES if AMI

Motor Control: Terminal knee control; avoid uncontrolled pivoting

Balance Training: Bilateral weight shift → bilateral stance

Proprioception: Early gentle joint position as effusion allows

Functional Training: ADL gait, step-to stairs

Sport-Specific Progression: None yet

Criteria to Progress: Full extension, SLR no lag, effusion controlled, flexion milestones met

Return to Running: Not in this phase

Return to Sport: Not in this phase

Outcome Measures: IKDC, Lysholm, pain NPRS, effusion grade, quad lag

Clinical Guidelines: Criterion-based ACLR rehab frameworks

Evidence: Early extension and quad activation strongly supported to reduce arthrofibrosis

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. ACLR rehab consensus literature.

RECORD: ACL Reconstruction — Intermediate/Late Strength

ID: KNEE-REHAB-02

Condition: ACL reconstruction

Rehabilitation Phase: Intermediate to late strength (approx 6 weeks-6 months)

Goals: Strength symmetry progress, single-leg control, aerobic base

Pain Management: Monitor delayed onset pain; adjust volume

Weight Bearing Status: Full

ROM Goals: Full symmetrical ROM

Strength Goals: Progress closed chain; hamstring carefully per graft; aim toward >80-90% LSI before advanced plyometrics

Motor Control: Single-leg squat quality; land mechanics foundation

Balance Training: Single-leg stance progressions

Proprioception: Perturbation training progressive

Functional Training: Step-downs, bridges, bike/swim

Sport-Specific Progression: Linear movement preparation late in window

Criteria to Progress: Strength LSI thresholds, hop test readiness, no effusion with load

Return to Running: Often around 3-4+ months if criteria met

Return to Sport: Not yet — build capacity

Outcome Measures: Isokinetic/HHD LSI, hop battery, ACL-RSI

Clinical Guidelines: Criteria-based progression

Evidence: Strength and hop symmetry associated with better RTS outcomes

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine. Grindem/Kyritsis RTS criteria literature.

RECORD: ACL Reconstruction — RTS Phase

ID: KNEE-REHAB-03

Condition: ACL reconstruction

Rehabilitation Phase: Return to sport (often 9-12+ months)

Goals: Pass RTS battery; graded COD/contact exposure; psychological readiness

Pain Management: Should be minimal with high loads

Weight Bearing Status: Full

ROM Goals: Full

Strength Goals: Quad LSI often ≥90% recommended in many protocols

Motor Control: Cut/land quality under fatigue

Balance Training: Sport-specific unstable/chaotic

Proprioception: Reactive agility

Functional Training: Full practice participation graded

Sport-Specific Progression: Noncontact → contact; partial → full play

Criteria to Progress: Time + criteria (strength, hops, movement, ACL-RSI)

Return to Running: Already achieved earlier

Return to Sport: Clearance shared decision with surgeon/team

Outcome Measures: RTS test battery, ACL-RSI, reinjury tracking

Clinical Guidelines: Do not use time alone

Evidence: Passing criteria reduces but does not eliminate second injury risk

References: RTS criteria papers; Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Meniscal Repair Rehab

ID: KNEE-REHAB-04

Condition: Meniscal repair

Rehabilitation Phase: Protected healing then progressive (protocol-specific 0-4/6 weeks restricted often)

Goals: Protect repair, limit shear/deep flexion early per surgeon

Pain Management: Effusion control

Weight Bearing Status: NWB/PWB/WBAT varies by tear type/repair — follow surgeon

ROM Goals: Often limited flexion early (e.g. 0-90) then progress

Strength Goals: Quad activation early; avoid early weighted deep squat

Motor Control: Gait restoration when WB allows

Balance Training: When WB allows

Proprioception: Progressive

Functional Training: ADL first

Sport-Specific Progression: Delayed vs meniscectomy

Criteria to Progress: Surgeon milestones + effusion/pain

Return to Running: Often 3-4+ months

Return to Sport: Often 4-6+ months

Outcome Measures: Pain, effusion, IKDC

Clinical Guidelines: Repair protection principles

Evidence: Protocols vary — communicate with surgeon essential

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: PFPS Rehab

ID: KNEE-REHAB-05

Condition: Patellofemoral pain

Rehabilitation Phase: Active rehabilitation (weeks to months)

Goals: Reduce pain, restore capacity, return to desired activity

Pain Management: Load adjustment; relative rest from aggravators not complete rest

Weight Bearing Status: Full as tolerated

ROM Goals: Full; address restrictions if present

Strength Goals: Quad + hip abductor/ER progressive resistance

Motor Control: Step-down and squat mechanics

Balance Training: Single-leg control

Proprioception: Task-specific

Functional Training: Stairs, sit-to-stand graded

Sport-Specific Progression: Run/jump volume graded with pain rules

Criteria to Progress: Pain stable with increased load

Return to Running: Walk-run with symptom monitoring

Return to Sport: When training loads match sport demands

Outcome Measures: AKPS/Kujala, NPRS, functional step-down

Clinical Guidelines: PFPS consensus — exercise first-line

Evidence: Strong for combined hip+knee exercise

References: BJSM PFPS consensus themes. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Patellar Tendinopathy Rehab

ID: KNEE-REHAB-06

Condition: Patellar tendinopathy

Rehabilitation Phase: Progressive tendon loading continuum

Goals: Improve load tolerance of tendon and kinetic chain

Pain Management: Isometrics for analgesia; pain-monitoring model (e.g. ≤3-5/10 acceptable if settles)

Weight Bearing Status: Full

ROM Goals: Ankle DF and hip mobility as needed

Strength Goals: Heavy slow resistance 3-4x/week then energy storage

Motor Control: Landing mechanics

Balance Training: Single-leg landing prep

Proprioception: Jump-land awareness

Functional Training: Decline squat progressions

Sport-Specific Progression: Jump counts graded

Criteria to Progress: Pain rules + strength + plyometric tolerance

Return to Running: When tendon tolerates impact volume

Return to Sport: When sport-specific energy storage tolerated repeatedly

Outcome Measures: VISA-P, pain map, hop/jump load

Clinical Guidelines: Cook & Purdam / Malliaras frameworks

Evidence: Progressive loading superior to passive-only approaches

References: Cook & Purdam; Malliaras. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: MCL Injury Rehab

ID: KNEE-REHAB-07

Condition: MCL sprain

Rehabilitation Phase: Protected motion to RTS (grade-dependent)

Goals: Allow ligament healing, restore ROM/strength, return without valgus apprehension

Pain Management: Brace comfort, ice early

Weight Bearing Status: WBAT usually in hinged brace

ROM Goals: Early ROM in brace; avoid forcing valgus

Strength Goals: Quad/hip; progressive frontal plane

Motor Control: Frontal knee control

Balance Training: Progressive

Proprioception: Progressive

Functional Training: Bike, swim, linear run before COD

Sport-Specific Progression: Shielded contact → full

Criteria to Progress: Painless valgus stress, strength, hop

Return to Running: When walking pain-free and brace plan allows

Return to Sport: Grade I weeks; III often 6-12+ weeks

Outcome Measures: Pain, laxity grade, function

Clinical Guidelines: Nonoperative MCL pathways

Evidence: Good healing with early motion for most

References: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

RECORD: Knee OA Rehab

ID: KNEE-REHAB-08

Condition: Knee osteoarthritis

Rehabilitation Phase: Long-term self-management with flares

Goals: Function, pain control, delay/optimize surgical timing if needed

Pain Management: Activity pacing, ice/heat, medications per physician, injections selective

Weight Bearing Status: Full; assistive device if helpful; unloader brace selected

ROM Goals: Maintain extension and functional flexion

Strength Goals: Quad and hip strength cornerstone

Motor Control: Sit-to-stand, stair strategies

Balance Training: Fall prevention

Proprioception: Stable stance tasks

Functional Training: Walking program, cycling

Sport-Specific Progression: Modify impact as needed

Criteria to Progress: Symptom-guided increases

Return to Running: Individual — many switch to lower impact; some tolerate graded run

Return to Sport: Participation goals individualized

Outcome Measures: WOMAC, KOOS, 30-sec chair stand

Clinical Guidelines: OARSI/NICE — education, exercise, weight loss first-line

Evidence: Strong for exercise and weight loss

References: OARSI/NICE OA themes. Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

13. Evidence and Guidelines

RECORD: Key Evidence and Guidelines — Knee Module

ID: KNEE-EVID-01

Topic: Foundational references for Kinora knee AI orientation

Anatomy: Standring S. Gray's Anatomy. Moore KL. Netter FH.

Biomechanics: Neumann DA. Kinesiology of the Musculoskeletal System.

Assessment: Magee DJ. Orthopedic Physical Assessment. Brukner & Khan Clinical Sports Medicine.

ACL: AAOS/AOSSM-aligned themes; criteria-based RTS literature (e.g. Grindem, Kyritsis, ACL-RSI).

Meniscus: Meniscus preservation and exercise-first evidence for many degenerative tears.

PFPS: BJSM consensus themes — hip + knee exercise, education, load management.

Tendinopathy: Cook & Purdam continuum; Malliaras progressive loading; VISA-P.

OA: OARSI / NICE osteoarthritis guideline themes — exercise and weight loss first-line.

Vascular: Knee dislocation — urgent vascular assessment protocols.

AI Use Note: ROM values, force multiples, and timelines are approximate and protocol-dependent. Prefer current surgeon protocols and individual assessment.

References: See fields above.